Solution: The Cartesian product $A \times B \times C$ consists of all ordered triples (a,b,c), where $a \in A$, $b \in B$, and $c \in C$.

$\therefore A \times B \times C = \{(0,1,0),(0,1,1),(0,1,2),(0,2,0),(0,2,1),(0,2,2),(1,1,0),(1,1,1),(1,1,2),(1,2,0),(1,2,1),(1,2,2)\}$ □

- More generally, $A^n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \ldots, n\}$

Example 20: Suppose that $A = \{1,2\}$. $A^2 = \{(1,1),(1,2),(2,1),(2,2)\}$

$A^3 = \{(1,1,1),(1,1,2),(1,2,1),(1,2,2),(2,1,1),(2,1,2),(2,2,1),(2,2,2)\}$ □

A subset R of the Cartesian product $A \times B$ is called a **relation** from the set A to the set B. The elements of R are ordered pairs, where the first element belongs to A and the second to B. For example, $R = \{(a,0),(a,1),(a,3),(b,1),(b,2),(c,0),(c,3)\}$ is a relation from the set $\{a,b,c\}$ to the set $\{0,1,2,3\}$, and it is also a relation from the set $\{a,b,c,d,e\}$ to the set $\{0,1,3,4\}$.

A relation from a set A to itself is called a relation on A.

Example 21: What are the ordered pairs in the less than or equal to relation, which contains (a,b) if $a \leq b$, on the set $\{0,1,2,3\}$?

Solution: $R = \{(0,0),(0,1),(0,2),(0,3),(1,1),(1,2),(1,3),(2,2),(2,3),(3,3)\}$ □

2.1.8 Truth Sets and Quantifiers

- Given a predicate P, and a domain D, we define the **truth set** of P to be the set of elements x in D for which $P(x)$ is true. The truth set of $P(x)$ is denoted by $\{x \in D \mid P(x)\}$.

Example 23: What are the truth sets of the predicates $P(x)$, $Q(x)$ and $R(x)$, where the domain is the set of integers and $P(x)$ is "$|x|=1$," $Q(x)$ is "$x^2 = 2$," and $R(x)$ is "$|x| = x$."

Solution: The truth set of P is $\{-1,1\}$. The truth set of R is the set of all natural numbers, i.e. N. It's easy to see that the truth set of Q must be an empty set.